Exploring Collaboration between a Language and a Non-Language Agent

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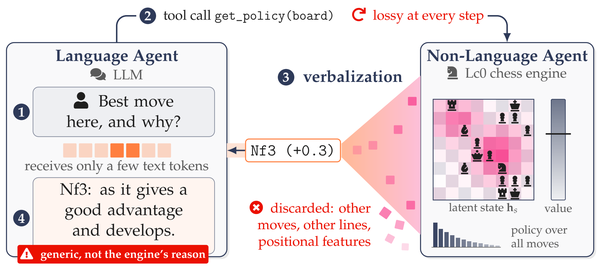
Motivation

How can an LLM effectively collaborate with non-language agents?

- LLMs have become general purpose orchestrators that can solve complex domain specific tasks by coordinating with tools and domain expert subagents.
- However, in many important domains such as robotics and game playing, the strongest available agents are **not language models**. Their expertise lives in rich internal representations (e.g. AlphaZero, RT-1).
- We study this collaboration using **chess** as a controlled environment owing to strong open engines (Lc0), public datasets, and real world tasks.

Verbalization Is Lossy

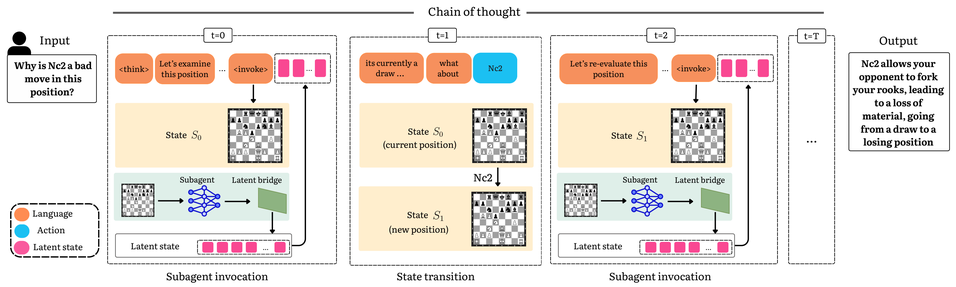
- Current methods call the agent as a tool and **verbalize** its output into text. However, this collapses the rich internal representation, into a few text tokens. We show the first quantification of this **Verbalization Debt**.



Our Contributions

- **Verbalization Debt:** the first quantification of what is lost when a non-language agent's knowledge is squeezed into text.
- **Llamia:** a language agent that internalizes a frozen non-language agent's latent state into its chain of thought, trained with RL.
- **Llamia-Bench:** six chess tasks over human behavior and positions, including three new datasets.
- A single Llamia-14B is **best on all six tasks**, learns **task-adaptive collaboration strategies**, and is preferred by skilled human players.

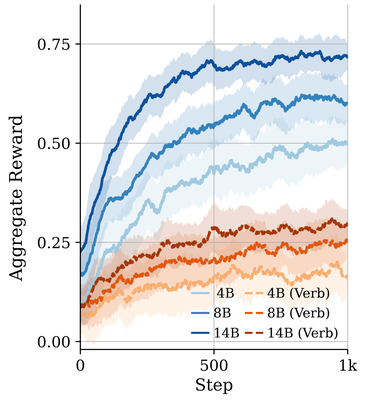
Latent State Internalization



- **Setup.** A **language agent** (Qwen3 LLM) collaborates with a frozen **non-language agent** (Lc0-BT4 chess engine); its internal state is projected into the LLM's chain of thought instead of being verbalized.
- **Projection.** Lc0's hidden state $\mathbf{h}_s$ (layer 14 of 15) is mapped by a 3-layer MLP, the **Latent-Bridge**, into the LLM's CoT at each invocation as k=32 **state tokens**: $\mathbf{z}_s = H_\varphi(\mathbf{h}_s)$.
- **Dynamic invocation.** Inside its **language** chain of thought, the LLM can repeatedly decide **when** to call the subagent and execute **actions** that update the board to counterfactual states, and call `get_policy` on any position. Each call re-encodes the current state, so the LLM always reasons over the position it is analyzing.

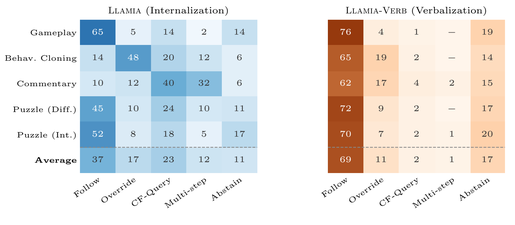
Training with Reinforcement Learning

- **Stage 1: projector alignment.** We follow LLaVA style two stage training, training only the LatentBridge on large scale synthetic state–value pairs.
- **Stage 2: reinforcement learning.** We post-train the LLM and LatentBridge jointly with **DAPO** on six of LLAMIA-Bench tasks. **State tokens** are masked from the policy gradient. The engine stays frozen in both stages. The resulting model, **Llamia**, reasons over **language**, **action** and **state** tokens.
- **Verbalization Baseline:** We also train Llamia-Verb replacing the state tokens with verbalized summaries (best moves, evaluation, position)



Emergent collaboration strategies

- We annotate the LLM and Subagent collaboration patterns using GPT-4o e.g. Follow, where the LLM simply follows the engine.
- Llamia **adapts its strategy to the task**: engine-follow for gameplay, consult-then-override for behavior cloning, counterfactual queries for commentary.
- Llamia-Verb **collapses to engine-follow** on every task (62–76%).



Strategy share (%) over 500 episodes per task (GPT-4o judge, κ=0.78). Left: Llamia. Right: Llamia-Verb.

Llamia-Bench

- We introduce **LLAMIA-Bench** a compilation of six popular chess tasks including Behavior Cloning, Move/Game Commentary Generation, and Puzzle Understanding including three new datasets
- None of these tasks can be solved by pretrained LLMs or Engines alone.

Behavior Cloning — Elo 1500 → Move a 1500-rated human would play: Nf3 (12M games, 5 Elo bands)

Puzzle Difficulty — + solution → Difficulty for human solvers: 1850 (4M Lichess puzzles)

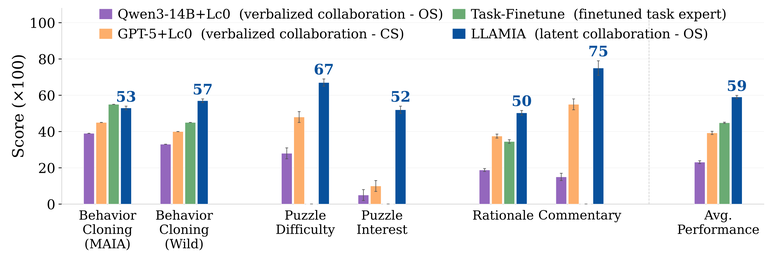
Move Rationale — move Bb5 → Why did the player make this move (90K annotated games)

BC in the Wild NEW — + context → Same, for GMs, low clock, rating gaps: Qxd5 (GM, low-time, Elo-gap games)

Puzzle Interest NEW — + solution → How much players liked it: +72 votes (no text proxy exists)

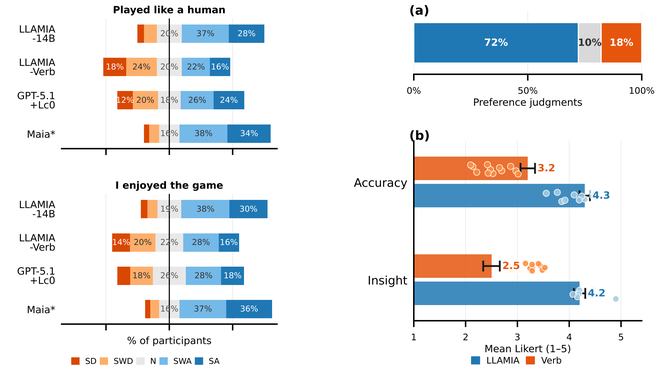
Game Commentary NEW — full game → Commentate on this game (Agadmator-2K: 1.9K games)

Results



- A single Llamia-14B model is best on all six tasks, across all tasks and sizes, ahead of GPT-5 with the same engine as a tool, and even existing state of the art task-specific finetuned models.
- **Joint reasoning needs latent state.** On harder tasks that require reasoning jointly over human behavior and the position, such as Puzzle Interest, only Llamia scores well; verbalization catches up only on tasks a text summary can approximate, such as Puzzle Difficulty.
- **Beyond chess.** On Go with KataGo, the same method is ~10 points above the text-based control.

Human study



- **Human study** (n=12 players rated ≥1700): post-game ratings put Llamia on par with Maia* for human-likeness; 72% of commentary judgments prefer Llamia.